Hanjin Gong

Richard Gong · Senior Software Engineer

Financial Systems · Backend & Full-Stack Development

Tokyo, Japan · Open to relocation

Email · LinkedIn · GitHub

Profile

Senior Software Engineer with 18 years of experience across investment banking, retail banking, payments and consumer platforms. My background spans financial systems, backend and full-stack development, quantitative analytics, enterprise integration and deployment automation. At Nomura, I have worked on fixed-income portfolio and risk analytics, securities lending, cash operations and trading workflows.

I work across a broad technology stack, including Java, Python, TypeScript/JavaScript, Go, C# and Rust, alongside frameworks and tools such as NestJS, Next.js, Angular, SQL Server, QuantLib, Docker, GitLab CI/CD and Ansible. My interests also extend to AI, automated reasoning, intelligent agents, cloud computing and distributed systems, informed by postgraduate study and independent exploration of technologies such as Jason/AgentSpeak and Prolog.

I enjoy solving complex engineering problems and connecting financial-domain requirements with practical, maintainable software.

Professional experience

Senior Engineer | Nomura Securities

Contract assignment through Michael Page Japan

Tokyo, Japan · June 2022 - Present

Investment management, trading operations and financial analytics platforms

- Work on a portfolio of internal financial applications supporting fixed-income portfolio construction, portfolio and benchmark analysis, securities lending, cash operations, order simulation, and downstream trading-system integration.
- Develop and maintain backend and full-stack workflows spanning TypeScript/Node.js services, Next.js and Angular interfaces, Java applications, Python batch processing, and SQL Server-backed business data.
- Support portfolio and risk analysis covering positions, market data, yield curves, pricing, duration and key-rate sensitivities, factor exposures, and benchmark-relative aggregation.
- Contribute to Python-based financial processing for fixed-income analytics and portfolio/order-generation workflows, incorporating reusable calculation components, data-validation logic, configurable processing stages, and regression testing.
- Work on multi-asset order and trade workflows, including validation, permission checks, simulation states, transaction consistency, file-based downstream interfaces, and integration with established investment-management systems.
- Support operational applications for securities lending and fund cash management, bringing together dated position snapshots, broker allocation flows, cash forecasting, subscriptions/redemptions, FX activity, and exception handling.
- Develop and maintain enterprise deployment automation using GitLab CI/CD, Jenkins and Ansible across Linux and Windows environments, with environment-specific configuration, controlled releases, health checks, deployment logging, and service-lifecycle operations.

- Address integration and reliability challenges involving Windows authentication, legacy application compatibility, database/file consistency, financial-data quality, long-running batch processing, and production diagnostics.

Professional technology exposure: TypeScript/JavaScript · Node.js / NestJS · Next.js · Angular · Python · QuantLib · pandas · Java · SQL Server · TypeORM · REST APIs · Windows authentication · GitLab CI/CD · Jenkins · Ansible · Docker · Linux · Windows Server

The descriptions above summarise engineering areas across several internal systems. Proprietary project names, implementation identifiers and operational details are intentionally omitted.

Senior Software Engineer - Nisshin

Tokyo, Japan · Jan 2021 - Jun 2022

Platform: Talent Management System (HR SaaS)

- Built REST APIs on Java 17 + Javalin with a React / TypeScript front end for real-time HR analytics; enforced strict layering (handlers to services to repositories) to keep the codebase testable as the team grew.
- Introduced a Redis-backed caching strategy with TTL + cache-aside invalidation and rewrote N+1 MySQL queries into batched joins - cut median page-load time by ~30% and removed a recurring DB CPU spike during morning login peaks.
- Developed load and stress-testing scripts in Python and Go to evaluate application performance under concurrent workloads and support performance troubleshooting.
- Stood up GitLab CI/CD pipelines (build, test, container scan, staging deploy), mentored 5+ junior engineers on code review, branching model, and observability.

Stack: Java 17 · Javalin · Python · Go · React · TypeScript · MySQL · Redis · GitLab CI · Load & stress testing

Technical Leader - HSBC Bank

Guangzhou, China · Jan 2019 - Jan 2021

Platform: Overseas Bank Account Opening (30+ jurisdictions)

- Led architecture for a Java + Node.js microservices estate handling cross-border KYC, document capture, and account provisioning across 30+ countries with per-region compliance rules.
- Drove workflow automation (orchestration + retries + dead-letter handling) that cut manual operational overhead by ~20% and shortened average account-opening turnaround meaningfully.
- Rolled out Datadog observability across services - SLOs, latency histograms, error-budget alerting - replacing ad-hoc log grepping with actionable dashboards for on-call engineers.

Stack: Java · Spring Boot · Node.js · Oracle · MySQL · Kafka · Docker · Datadog

Technical Leader - SurLead

Manila, Philippines · Mar 2015 - Jan 2019

- Delivered FinTech and e-commerce client platforms in Java + TypeScript on Oracle / MySQL, owning architecture, delivery, and team leadership end-to-end.
- Migrated legacy deployments to Docker and split a monolith into NestJS / Express microservices - shortened release cycles by ~40% and enabled independent team ownership per service.

Stack: Java · TypeScript · NestJS · Express · Oracle · MySQL · Docker

Full-Stack Engineer - Ctrip

Shanghai, China · Jun 2013 - Feb 2015

Platform: Ctrip Wireless (travel-booking platform, millions of DAU)

- Co-designed the frontend framework (Node.js · Backbone.js · jQuery) used across booking, payment, and loyalty modules - standardised component patterns across product lines.
- Shipped user-facing features on the booking and payment critical path, serving millions of daily users with tight SLAs during peak travel seasons.

Stack: Node.js · Backbone.js · jQuery · Java (backend integrations) · MySQL

Front-End Engineer - Qunshuo

Shanghai, China · Aug 2010 - May 2013

- Built enterprise web apps on Java / Spring / Hibernate / MySQL with contemporary JS frameworks; delivered a SAP-integrated childcare module on schedule.

Core technologies

- **Professional application development:** Java, Python and TypeScript/JavaScript; Node.js / NestJS, Next.js, Angular, REST APIs and server-side application integration.
- **Financial engineering:** Fixed-income pricing and risk analytics, QuantLib, pandas, NumPy, yield curves, duration/key-rate sensitivity, portfolio and benchmark analysis, order simulation and securities lending workflows.
- **Data and enterprise integration:** SQL Server, SQL, TypeORM, file-based trade interfaces, authentication and authorisation, legacy financial systems and data-validation pipelines.
- **Infrastructure and delivery:** GitLab CI/CD, Jenkins, Ansible, Docker, Linux/Windows deployment automation, logging and health monitoring.
- **Broader programming languages: Go (Golang), C#, Rust, Jason/AgentSpeak and Prolog** - spanning earlier professional work, independent engineering and academic study. These complement my Java, Python and TypeScript/JavaScript experience.

Artificial Intelligence, Reasoning & Cloud Computing

- **AI and intelligent agents (postgraduate study):** Deductive and practical reasoning, knowledge representation, BDI agent architectures, reactive and hybrid agents, multiagent communication and coordination, with Jason/AgentSpeak and Prolog.
- **Machine learning foundations:** Mathematical foundations of learning, optimisation and model evaluation, complemented by independent Rust-based machine-learning experiments.
- **Cloud computing and distributed systems (postgraduate study and engineering interests):** Cloud architecture concepts, distributed application design and the relationship between scalable services, deployment automation and operational reliability. Distinguish academic study from production delivery experience.

Independent Research

- **Algorithmic trading with cTrader and MetaTrader 5 (MT5):** Research automated trading strategies, signal generation, order execution, position sizing and risk management, using backtesting and forward evaluation to examine strategy behaviour.
- **Model training with Rust:** Explore Rust-based machine-learning workflows, including data preparation, model training and evaluation, with attention to computational performance and memory efficiency.

- **Trading systems with Rust:** Investigate trading-system design, including market-data processing, asynchronous execution, concurrency and order management.
- **Multiagent systems with Jason and Prolog:** Explore BDI agents, logic-based reasoning, knowledge representation, inter-agent communication and coordination for collaborative decision-making.

Education

MSc Computer Science | University of Liverpool, UK Expected 2027

- **Multiagent systems (ongoing):** Agent autonomy, reactive and proactive behaviour, BDI (beliefs, desires and intentions), deductive and practical reasoning, agent communication and coordination.
- **Logic and agent-oriented programming (ongoing):** Prolog facts, rules, queries and unification; Jason/AgentSpeak beliefs, goals, plans, events and inter-agent messaging.
- **Additional study:** Mathematical foundations for machine learning, including calculus and gradient descent; human-computer interaction and accessibility evaluation.

Associate Degree, Science & Technology | Shanghai College of Science & Technology 2007-2010

Certifications and languages

- Cisco Certified Technician (CCT) - Routing & Switching.
- Microsoft Certified Professional & Specialist (2016).
- AI, Business & the Future of Work - Lund University (2023).
- English (fluent) · Chinese (native).

Contact

Open to senior backend, full-stack and financial-software engineering opportunities, especially in trading, risk and market-data systems.

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